

Extended exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"

Exercise: Computing a broadened absorption spectrum with NEWTON-X

based on the online NEWTON-X tutorial

(https://vdv.dcf.mybluehost.me/nx/wp-content/uploads/2020/02/tutorial-2_2.pdf)

see also the [CP2K manual webpage](#)

1. Computing a broadened absorption spectrum using nuclear ensembles and the classical NEWTON-X program package

- Exemplary input and output files are given in the directory
/zfs/home/suphc356/TEACHING/TEACHING_1004D_WS202526/Inputs_Exercise5_NewtonX
- Start from a converged geometry and perform a vibrational analysis, printing the harmonic frequencies adding the keyword `&NAMD_PRINT` to the section `&VIBRATIONAL_ANALYSIS`. All other keywords and settings can be chosen as for a conventional vibrational analysis computation. `&VIBRATIONAL_ANALYSIS`

```
&PRINT
&NAMD_PRINT
&END NAMD_PRINT
&END PRINT
DX 0.001
&END VIBRATIONAL_ANALYSIS
```
- To run the thereon based computation with NEWTON-X, the CP2K files have to be provided in a directory named `JOB_AD`. The CP2K input file has to be named `cp2k.inp` and the XYZ coordinates have to be provided as external file, dubbed `coord.cp2k`. Further specifications regarding the employed executable and parallelization setup of CP2K can be provided within the input file `cp2k.par`, specifying e.g. the following number of MPI tasks (keyword `parallel`) and the name of the executable (keyword `exec`):

```
parallel = 16
exec = cp2k.psmmp
```
- The CP2K input file `cp2k.inp` thus has to specify the XYZ coordinates as

```
&COORD
@include coord.cp2k
&END COORD
```
- Furthermore, an `initqp_input` file has to be provided for NEWTON-X (see instructions in the NEWTON-X tutorial). Specifications for CP2K in the `initqp_input` file are the following:
 - The file comprising the normal modes of the CP2K frequency computation — for the above input provided as `XXX-VIBRATIONS-1.eig` — has to be specified as `file_nmodes = XXX-VIBRATIONS-1.eig`.

- The electronic structure program has to be specified as CP2K by defining `iprog = 10`.

An exemplary `initqp_input` is thus given as

```
&dat
nact = 2
iprog = 10
numat = 3
npoints = 200
file_geom = geom
file_nmodes = XXX-VIBRATIONS-1.eig
anh_f = 1
rescale = n
temp = 0
ics_flg = n
chk_e = 1
nis = 1
nfs = 11
kvert = 1
de = 100
prog = 14
iseed = 0
lvprt = 1
/
```

The keyword `numat` specifies the number of atoms. `npoints` specifies the number of initial geometries and velocities to be generated; for our purposes, a relative small number is sufficient. `nis` and `nfs` refer to the number of excited states that were computed with CP2K, thus in the above exemplary input, the corresponding CP2K computation would have specified `NSTATES 10` in the `&TDDFPT` section to compute 10 excited states (+ the ground state, giving in total 11 states to be provided to NEWTON-X). `prog` has to be set equal to 14 in order to refer to CP2K as third-party program that provides not only the vibrational frequencies, but also the excited-state energies and gradients.

- To generate initial conditions, the script `initcond.pl` of NEWTON-X has to be executed. Modify the bash script for slurm by adding the following lines:

```
module load gcc12-env/12.1.0
module load gcc/13.2.0
module load gsl/2.7.1
module load autoconf/2.69
module load automake/1.15.1
module load netlib-lapack/3.11.0
module load openblas
export NX=/zfs/home/suphc356/SOFTWARE/NEWTONX/newtonx_CS_ownfork_Nov2024/bin
export PATH=$PATH:NX
```

Furthermore, the command to be executed is now no longer the `mpirun` call to the CP2K executable, but the call for the `initcond.pl` of NEWTON-X,

```
/zfs/home/suphc356/SOFTWARE/NEWTONX/newtonx_CS_ownfork_Nov2024/bin/initcond.pl > initcond.out
```

- The generated initial conditions are summarized in external output files for each state, dubbed "`final_output_XXX`", comprising information on the various generated geometries and velocities.

- Based on the initial conditions, the broadened photoabsorption spectrum can be computed with the `nxinp` script, `nxinp < inp_for_spectrum` with the input options specified in the `inp_for_spectrum` file as 6

```
1
1
2-11
F
0
-1
Empty line
Empty line
Empty line
0.05
Empty line
Empty line
Empty line
Empty line
Empty line
Empty line
8
```

"Empty line" indicates an empty line in the input file.

- As outlined in section 2.7 of the cited NEWTON-X tutorial, the so-obtained output file `cross-section.dat` comprises the data points of the computed photoabsorption spectrum. It can e.g. be straight-forwardly be visualized with `gnuplot`
`> gnuplot`
`gnuplot> plot 'cross-section.dat' using 1:3 with lines`
 or you can write a Jupyter Notebook to visualize such type of input data.